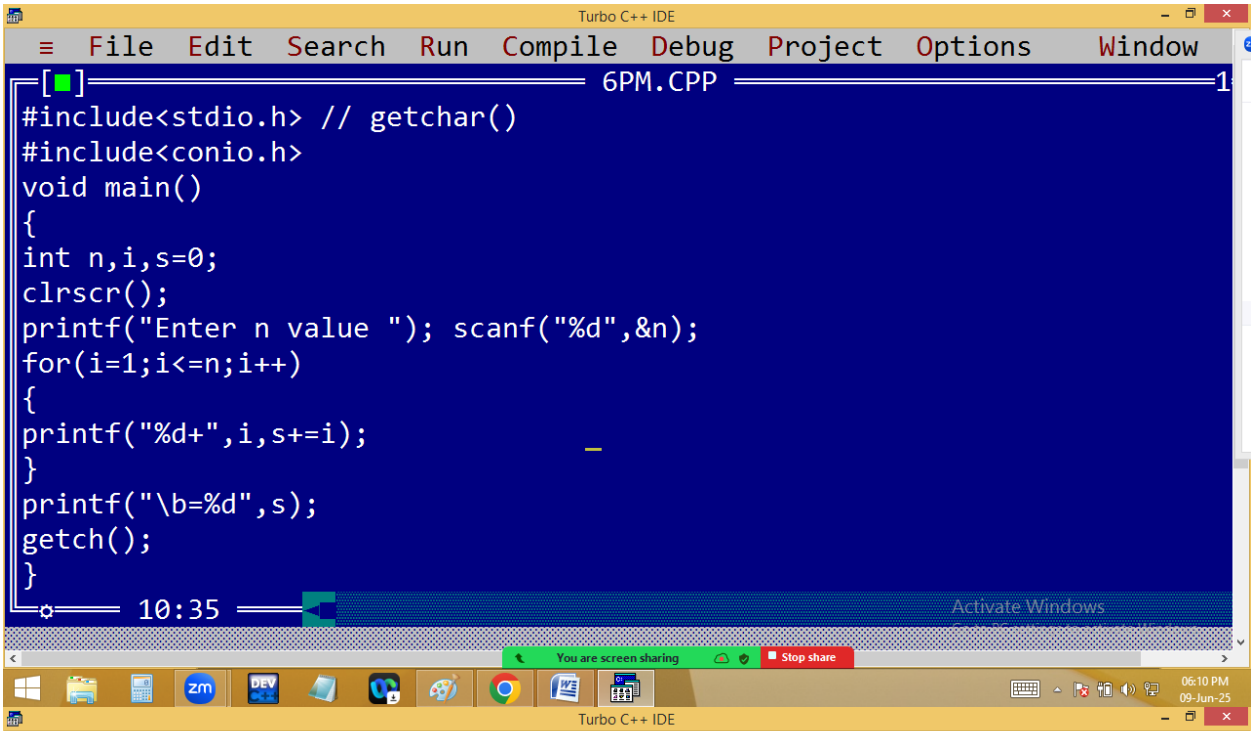
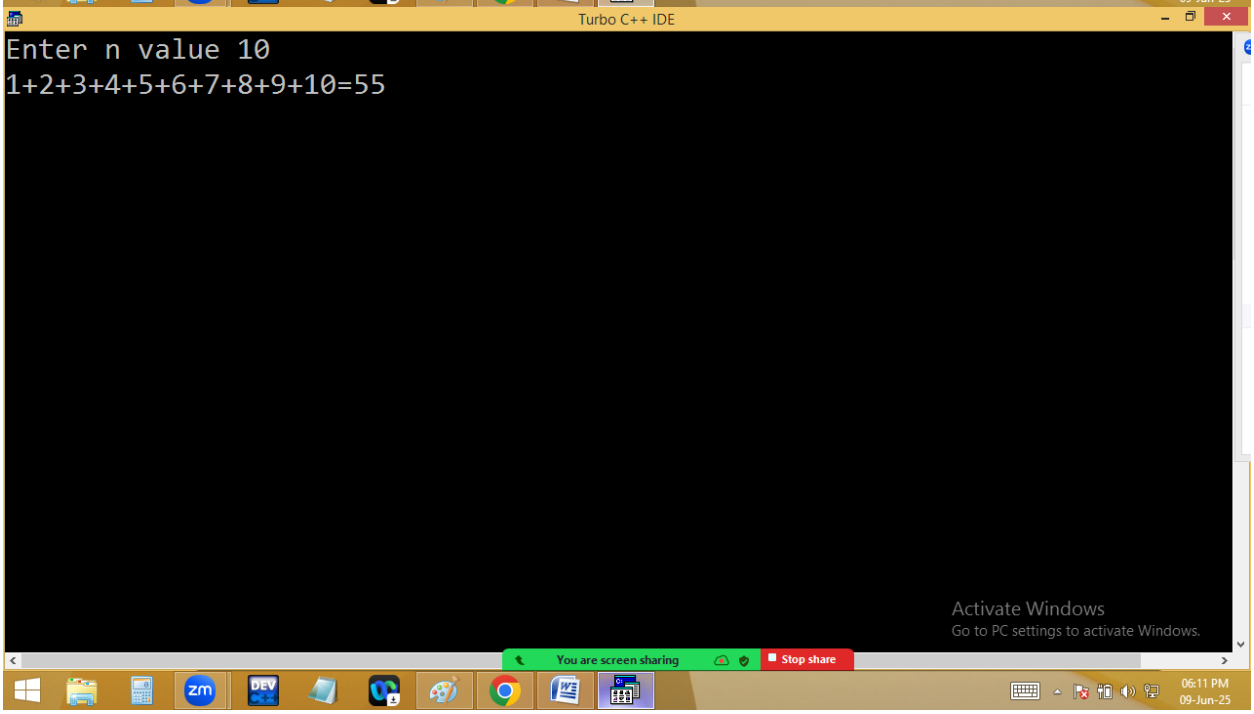
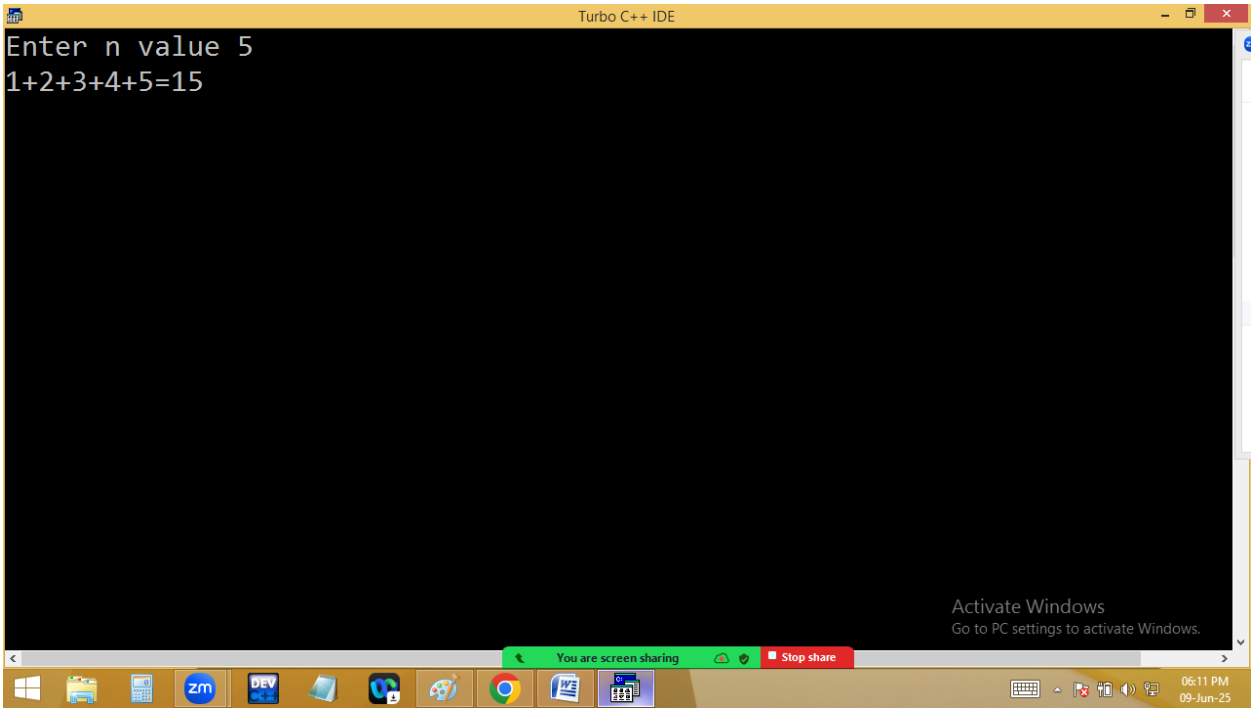
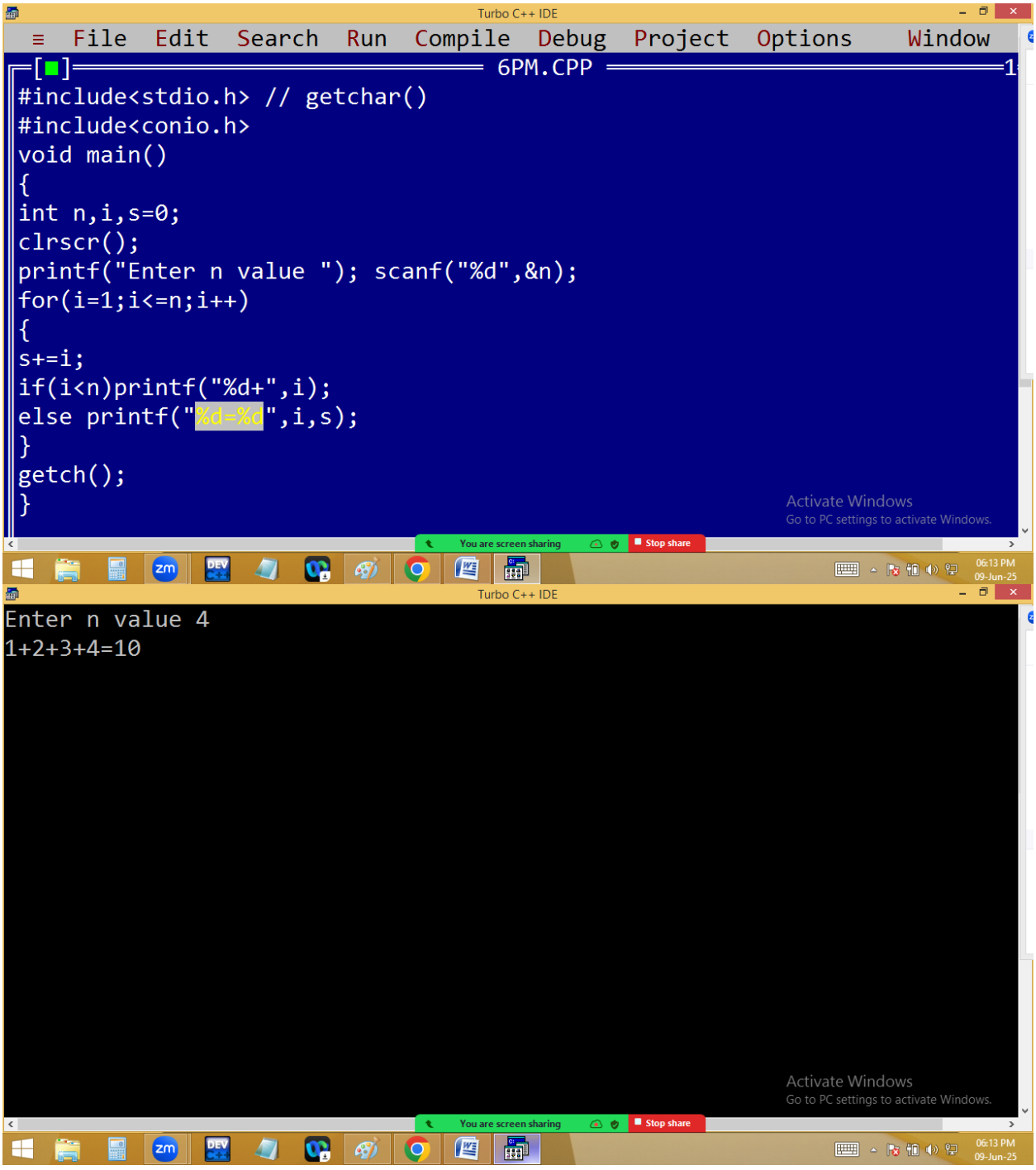
5 → 1+2+3+4+5=15


Turbo C++ IDE
File Edit Search Run Compile Debug Project Options Window
6PM.CPP
#include<stdio.h> // getchar()
#include<conio.h>
void main()
{
int n,i,s=0;
clrscr();
printf("Enter n value "); scanf("%d",&n);
for(i=1;i<=n;i++)
{
printf("%d+",i,s+=i);
}
printf("\b=%d",s);
getch();
}
10:35
Activate Windows
You are screen sharing
Stop share
06:10 PM
09-Jun-25

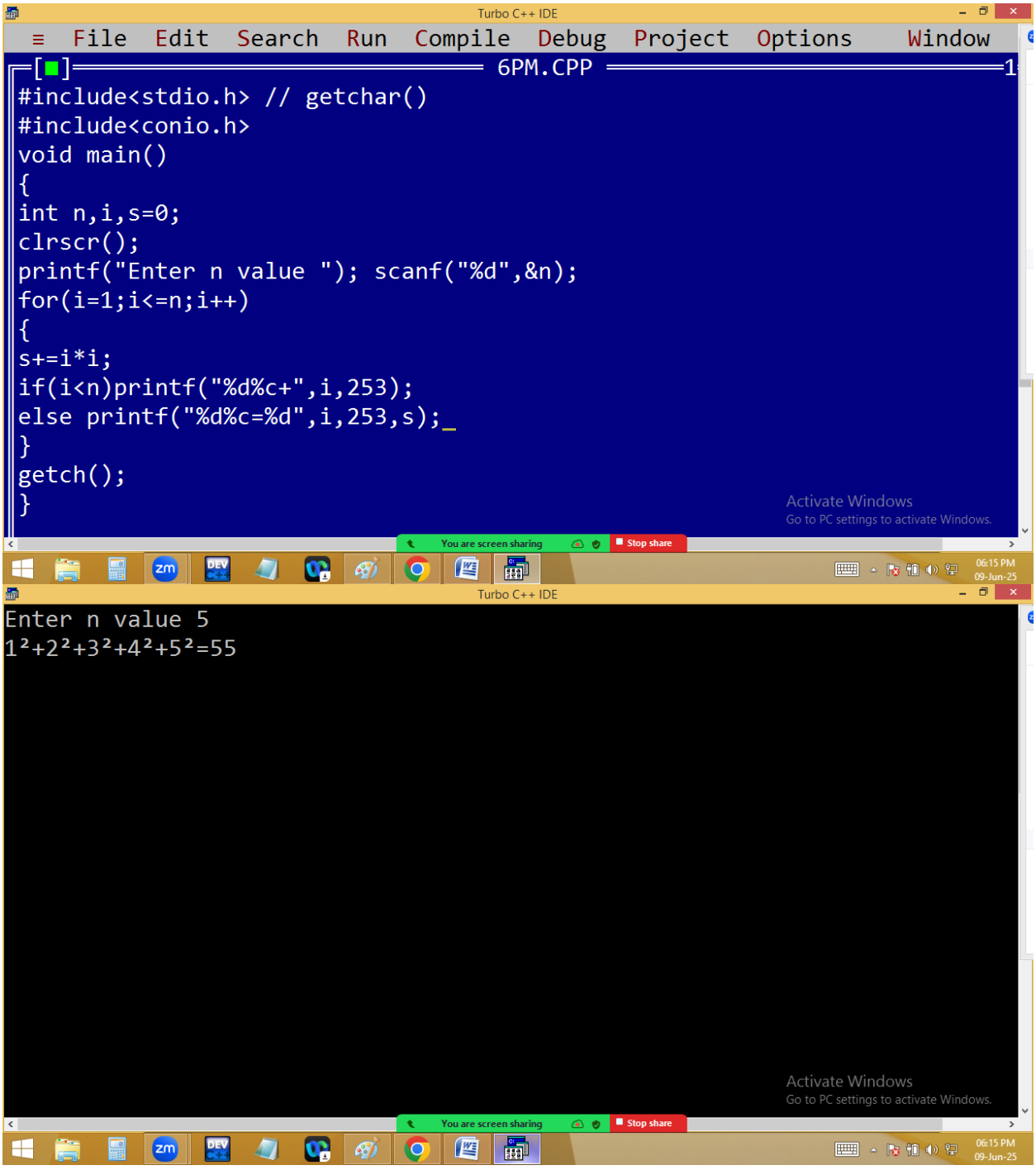

Turbo C++ IDE
Enter n value 10
1+2+3+4+5+6+7+8+9+10=55
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
06:11 PM
09-Jun-25



Without using \b:

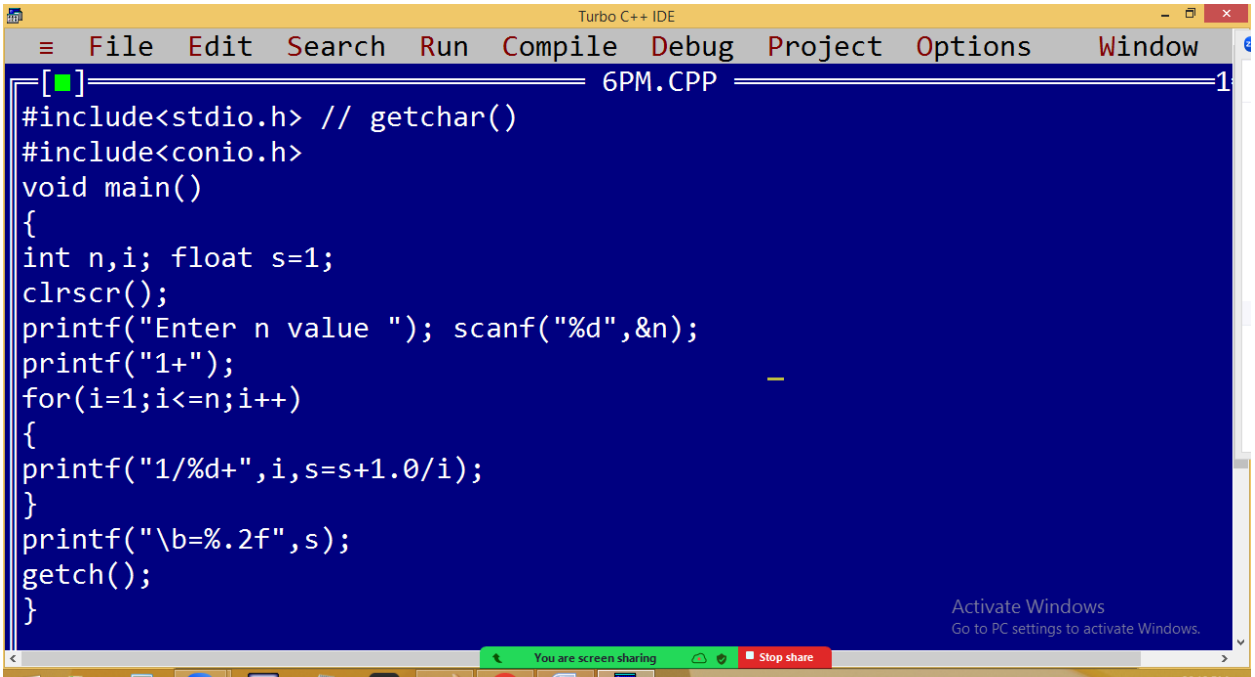


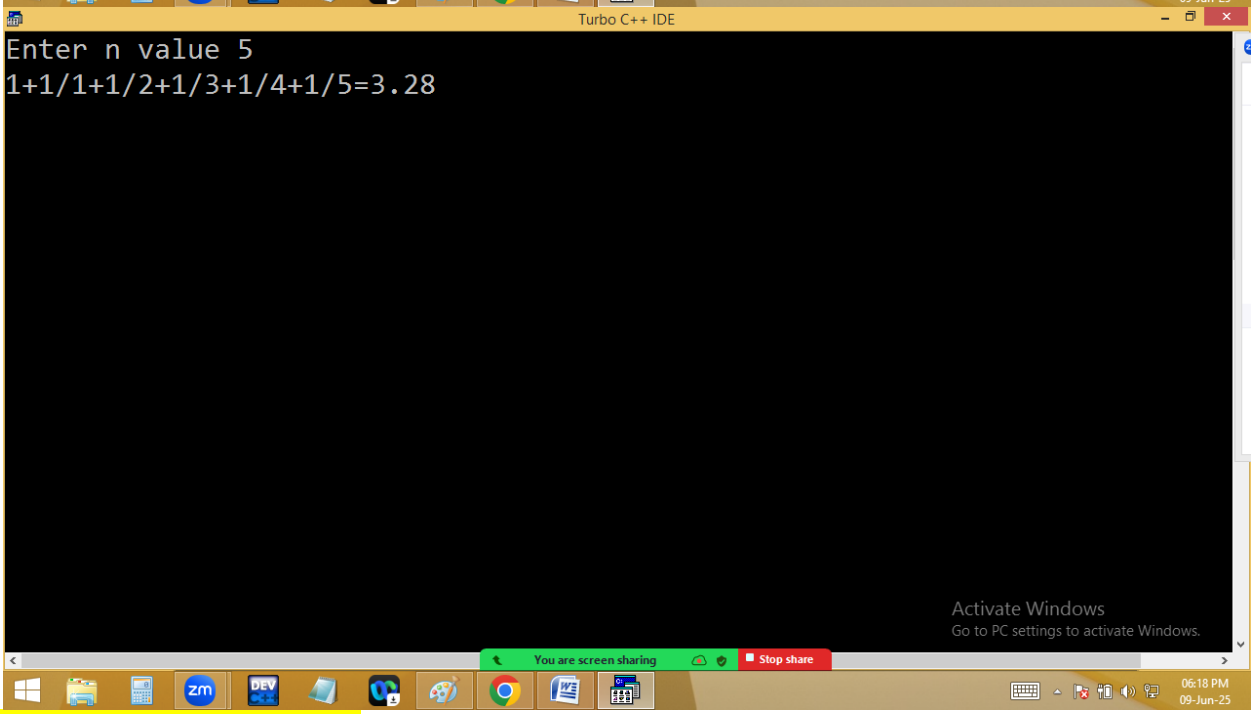
$5 \rightarrow 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$



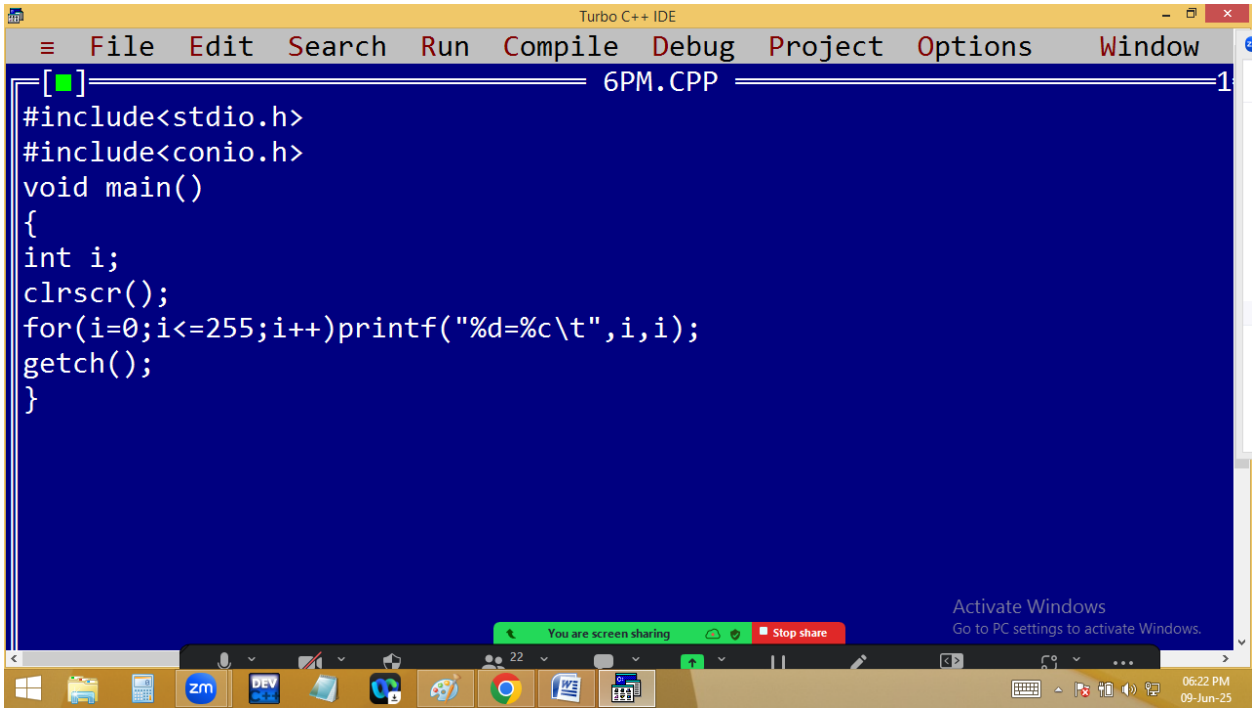
Harmonic series:

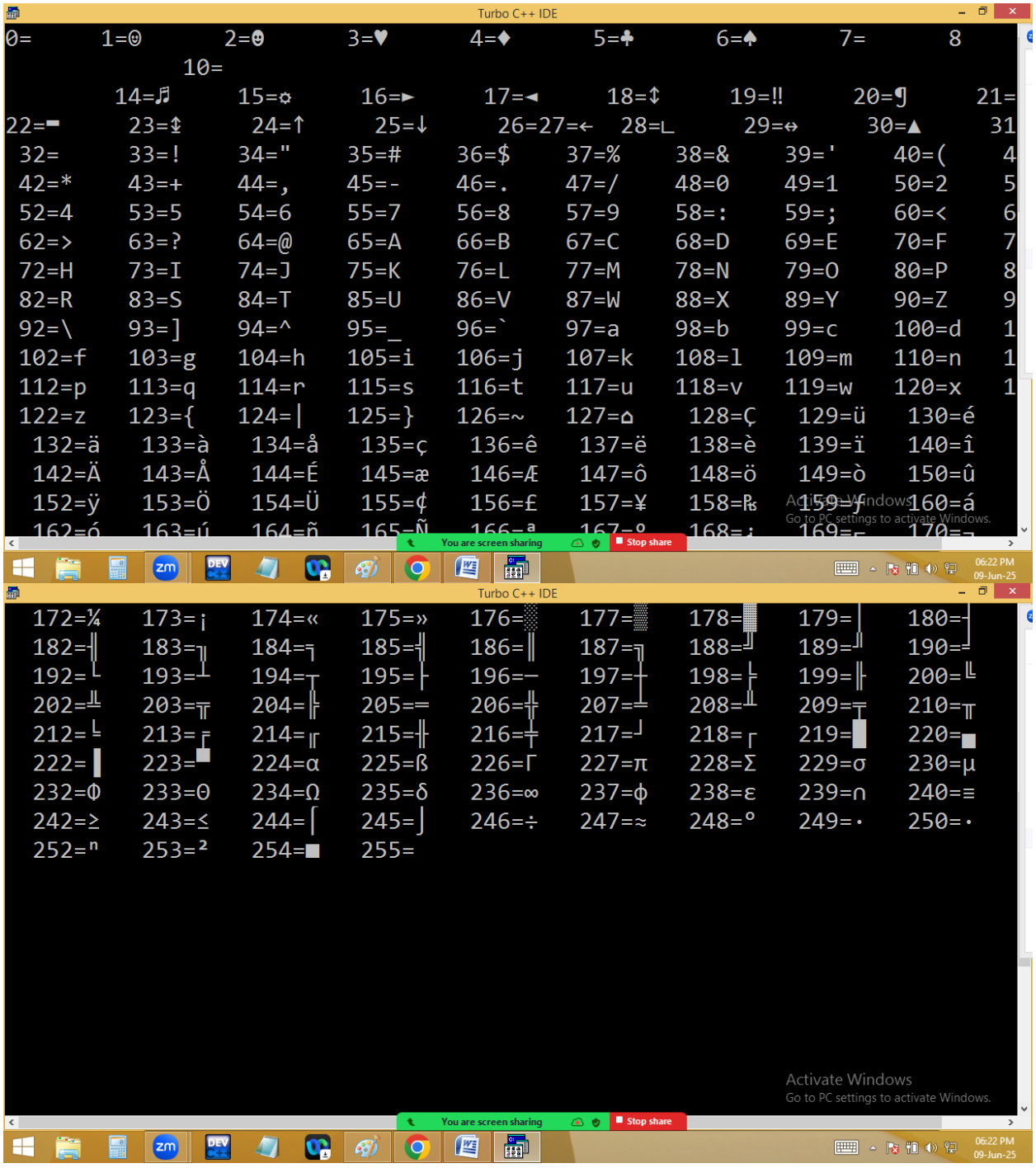
5 ➔ 1+1/1+1/2+1/3+1/4+1/5=3.28


Turbo C++ IDE
File Edit Search Run Compile Debug Project Options Window
6PM.CPP
#include<stdio.h> // getchar()
#include<conio.h>
void main()
{
int n,i; float s=1;
clrscr();
printf("Enter n value "); scanf("%d",&n);
printf("1+");
for(i=1;i<=n;i++)
{
printf("1/%d+",i,s=s+1.0/i);
}
printf("\b=%.2f",s);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
06:18 PM
09-Jun-25


Turbo C++ IDE
Enter n value 5
1+1/1+1/2+1/3+1/4+1/5=3.28
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
06:18 PM
09-Jun-25

ASCII TABLE:


Turbo C++ IDE
File Edit Search Run Compile Debug Project Options Window
6PM.CPP
#include<stdio.h>
#include<conio.h>
void main()
{
int i;
clrscr();
for(i=0;i<=255;i++)printf("%d=%c\t",i,i);
getch();
}
You are screen sharing
Stop share
Activate Windows
Go to PC settings to activate Windows.
06:22 PM
09-Jun-25



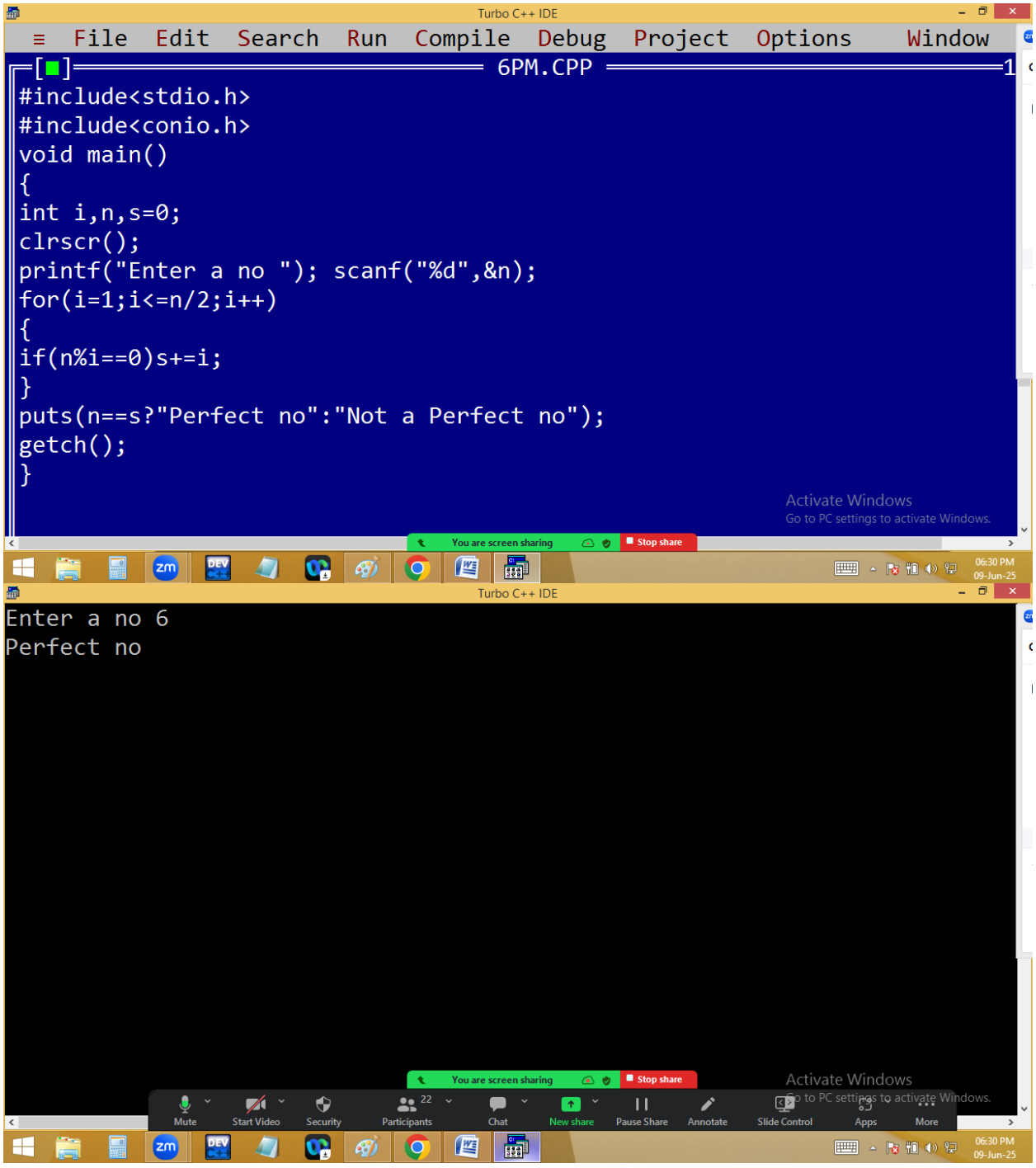
Finding perfect no or not?

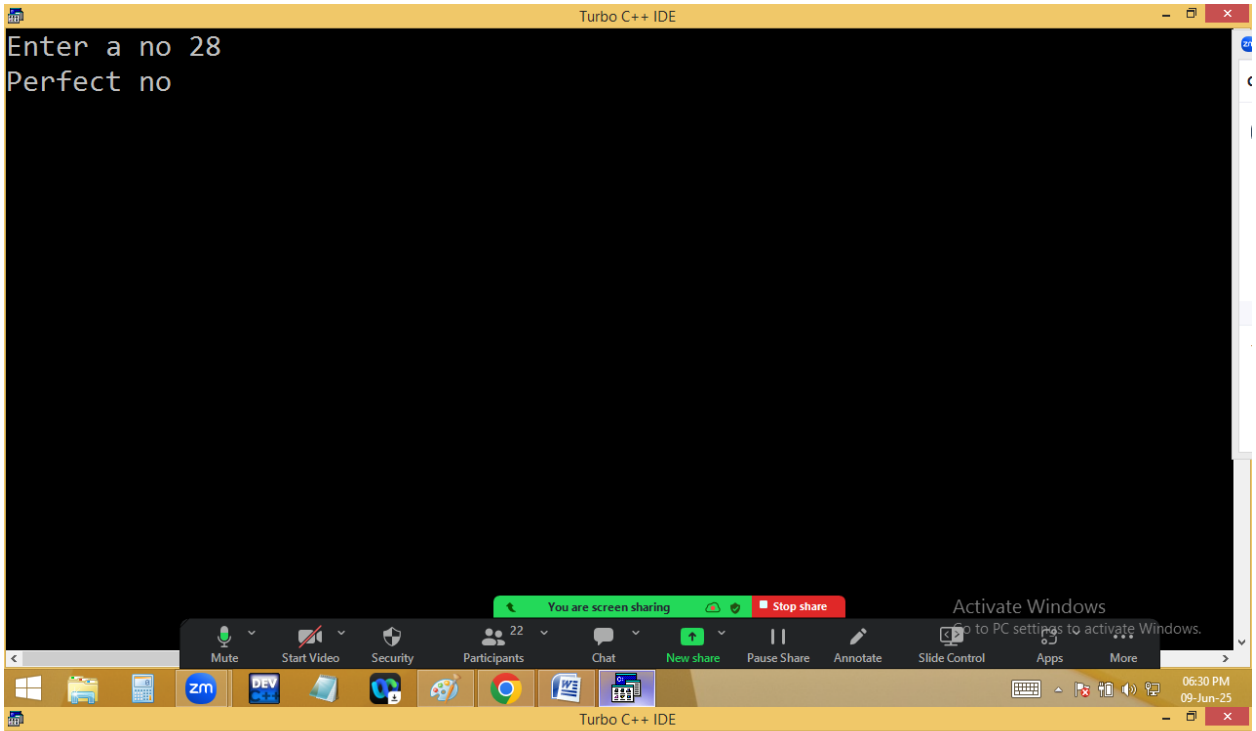
Sum of factors is equal to given no.

6 factors sum is 1+2+3=6

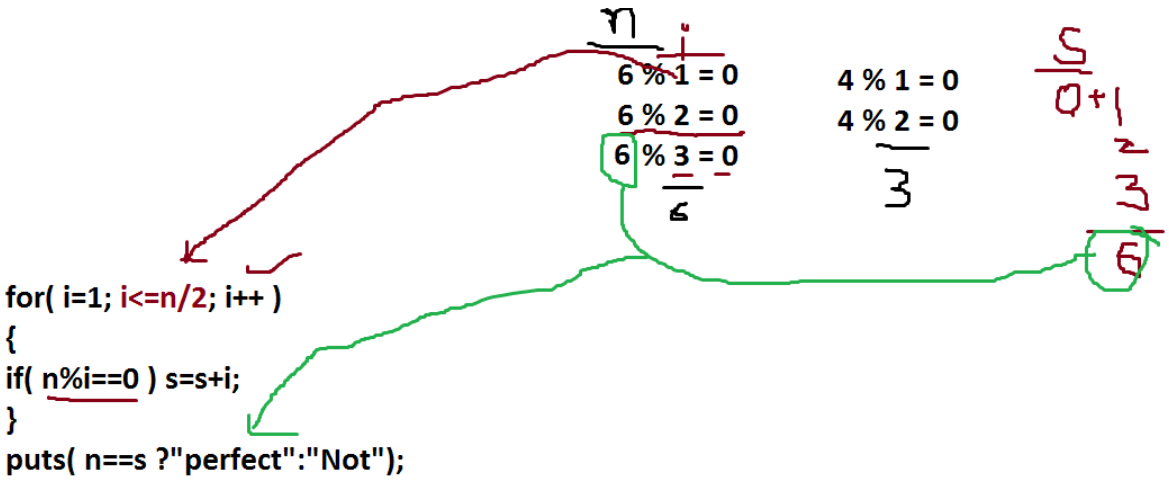
28 factors sum is 1+2+4+7+14=28

4 factors are 1+2=3 ← not a perfect no


Turbo C++ IDE
File Edit Search Run Compile Debug Project Options Window
6PM.CPP
#include<stdio.h>
#include<conio.h>
void main()
{
int i,n,s=0;
clrscr();
printf("Enter a no "); scanf("%d",&n);
for(i=1;i<=n/2;i++)
{
if(n%i==0)s+=i;
}
puts(n==s?"Perfect no":"Not a Perfect no");
getch();
}
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
Turbo C++ IDE
Enter a no 6
Perfect no
You are screen sharing
Stop share
Activate Windows
Go to PC settings to activate Windows.
Mute
Start Video
Security
Participants
Chat
New share
Pause Share
Annotate
Slide Control
Apps
More


Turbo C++ IDE
Enter a no 28
Perfect no
You are screen sharing
Stop share
Mute
Start Video
Security
Participants
22
Chat
New share
Pause Share
Annotate
Slide Control
Apps
More
Activate Windows
06:30 PM
09-Jun-25


Turbo C++ IDE
Enter a no 4
Not a Perfect no
You are screen sharing
Stop share
Mute
Start Video
Security
Participants
21
Chat
New share
Pause Share
Annotate
Slide Control
Apps
More
Activate Windows
06:30 PM
09-Jun-25



Finding prime/ composite no?

The no divisible with 1 and itself is a prime

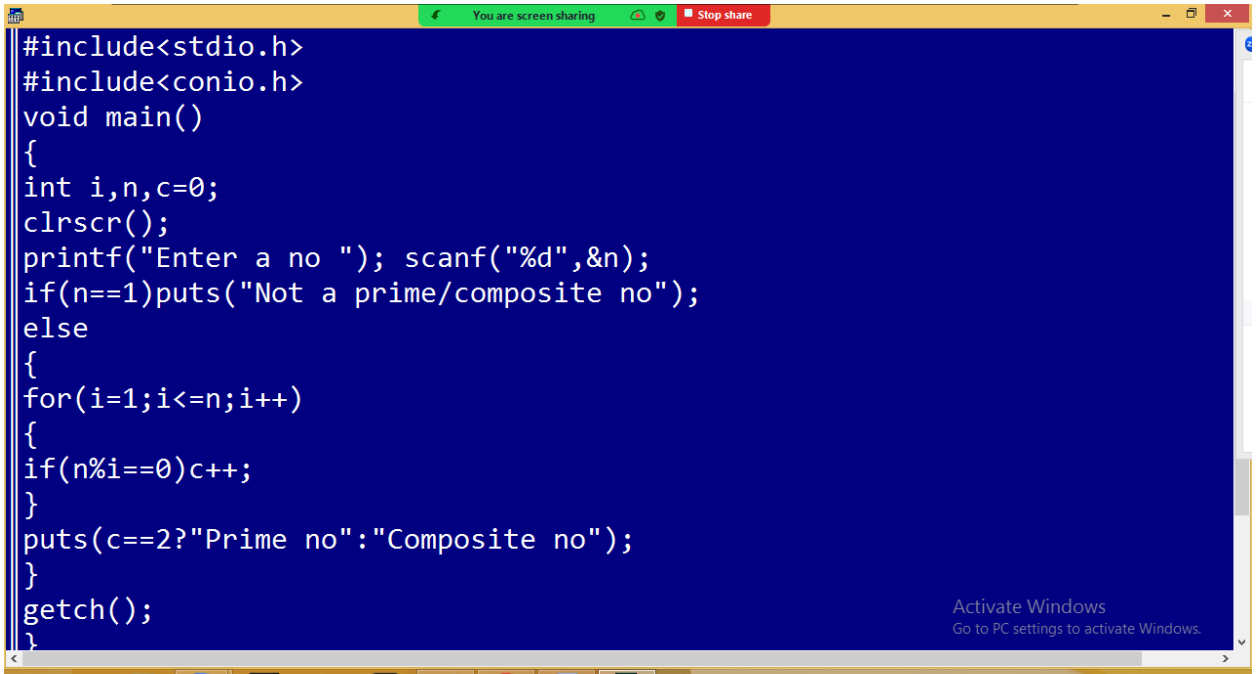
Or

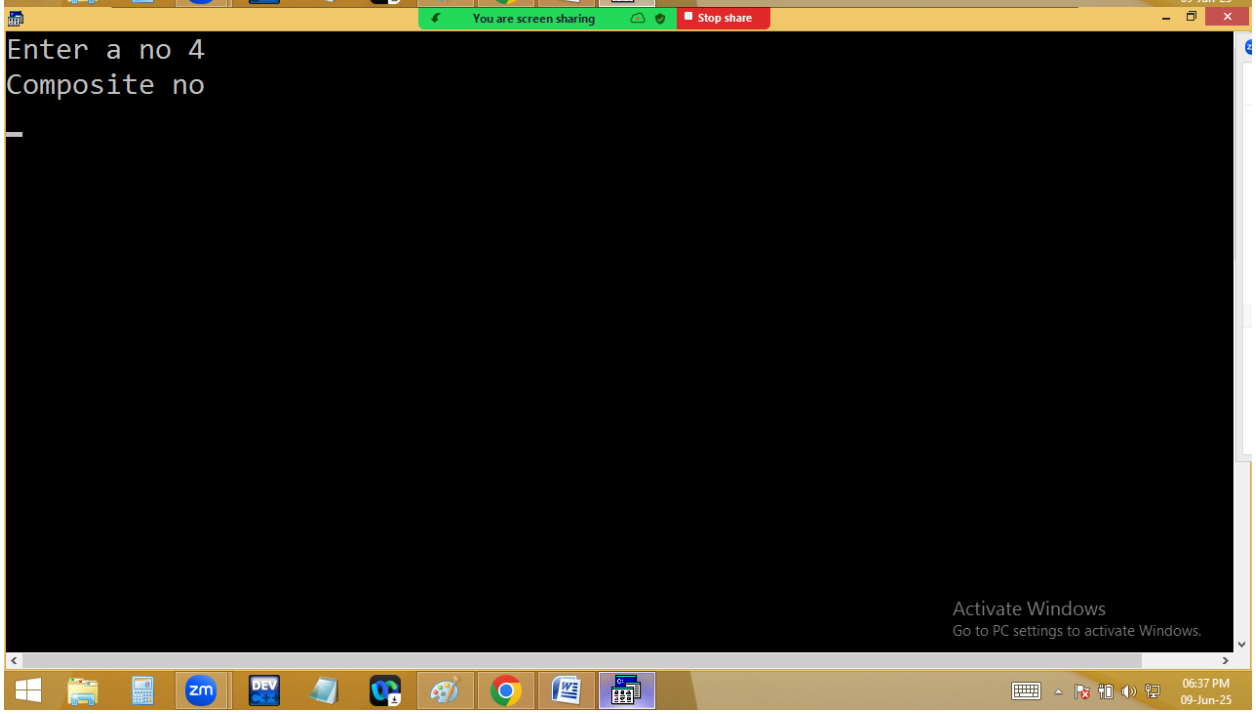
The no having 2 factors only is called prime

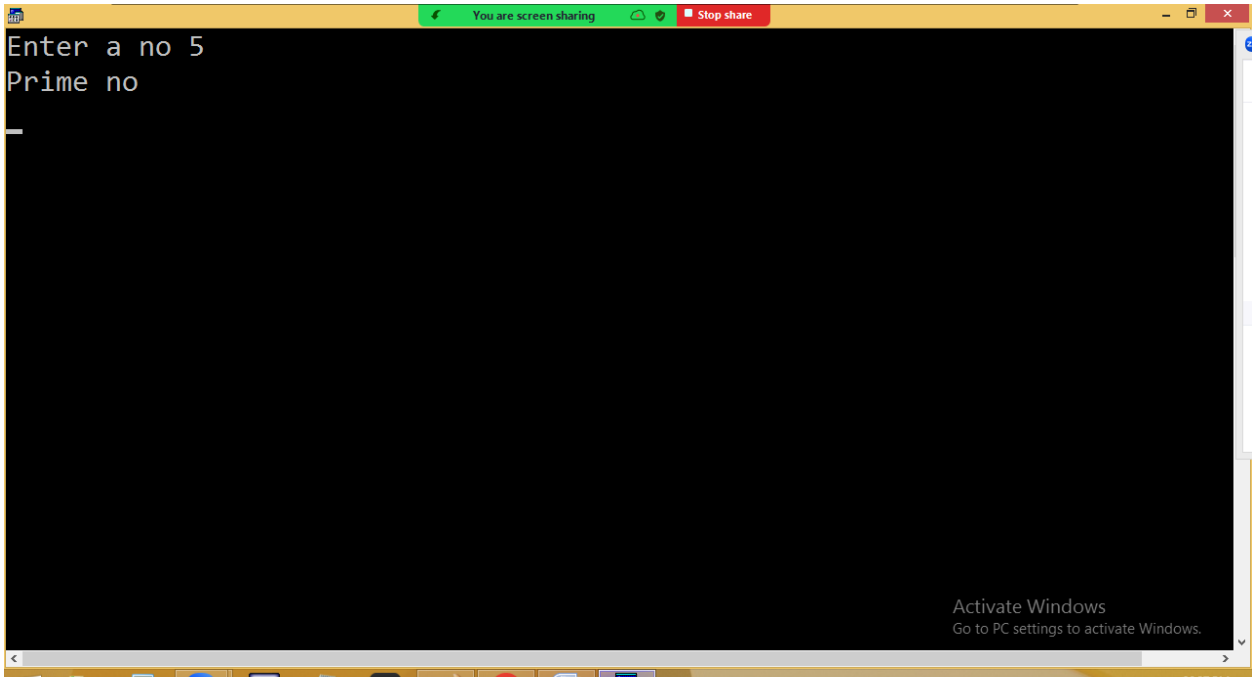
2 divisible with 1 and 2 only ← 2 factors ← prime

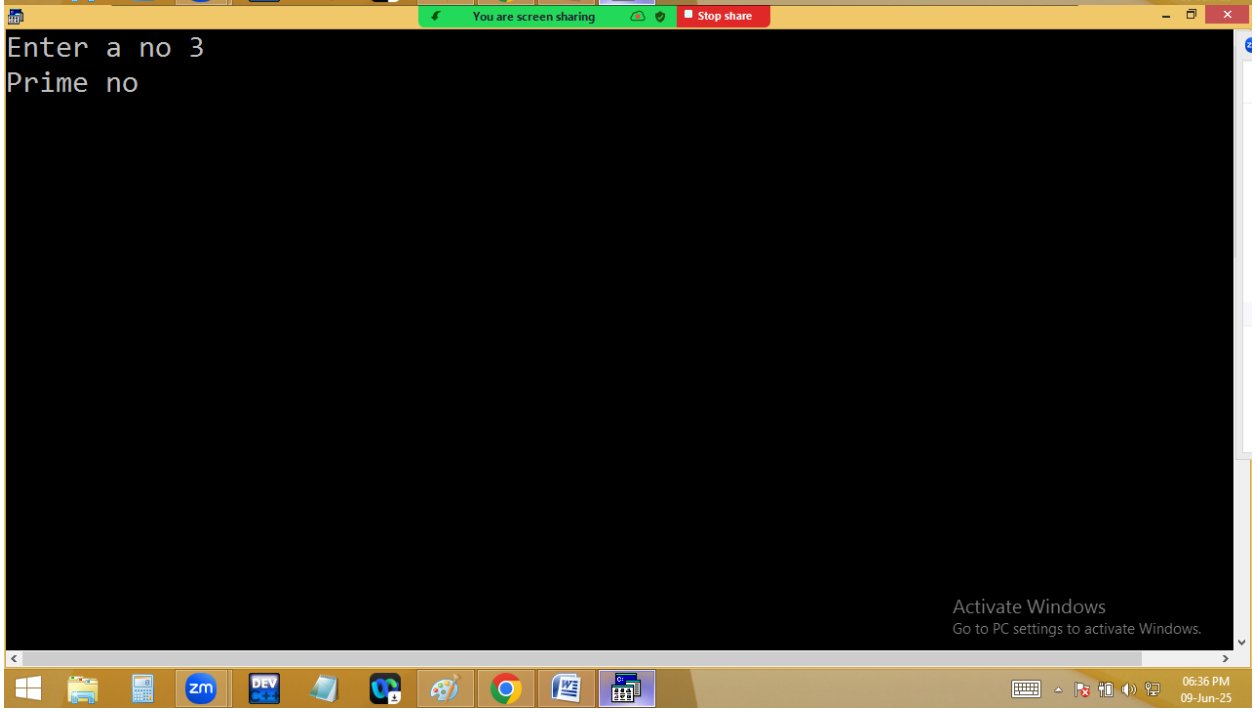
3 divisible with 1 and 3 only ← 2 factors ← prime

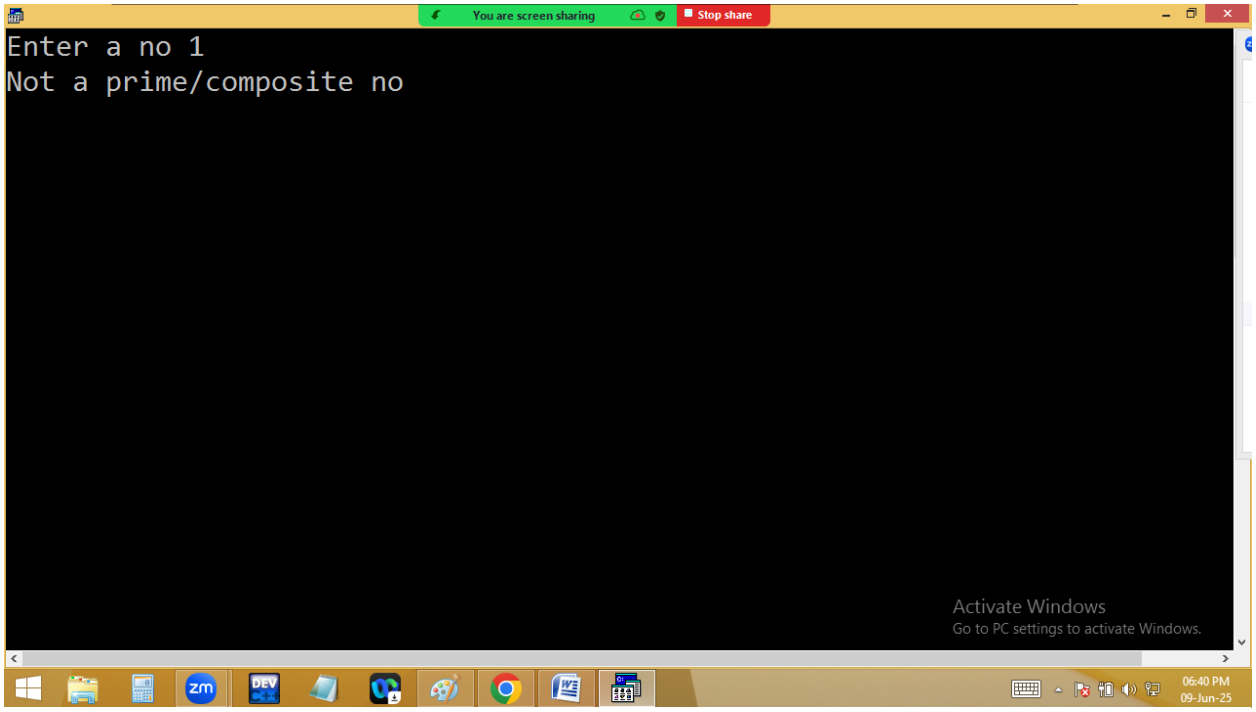
4 divisible with 1, 2, 4 ← 3 factors ← composite

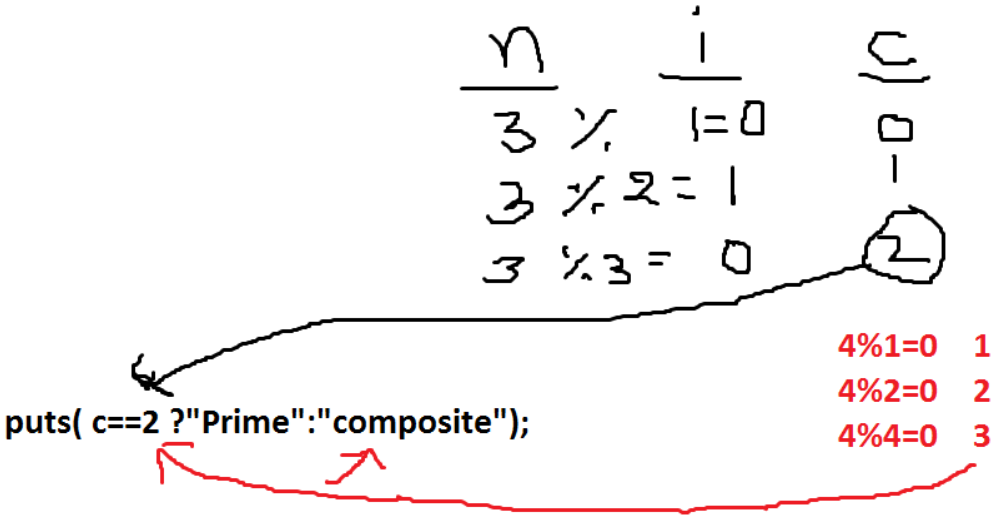

You are screen sharing
Stop share
#include<stdio.h>
#include<conio.h>
void main()
{
int i,n,c=0;
clrscr();
printf("Enter a no "); scanf("%d",&n);
if(n==1)puts("Not a prime/composite no");
else
{
for(i=1;i<=n;i++)
{
if(n%i==0)c++;
}
puts(c==2?"Prime no":"Composite no");
}
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:41 PM
09-Jun-25


You are screen sharing
Stop share
Enter a no 4
Composite no
Activate Windows
Go to PC settings to activate Windows.
06:37 PM
09-Jun-25


You are screen sharing
Stop share
Enter a no 5
Prime no
Activate Windows
Go to PC settings to activate Windows.
06:37 PM
09-Jun-25

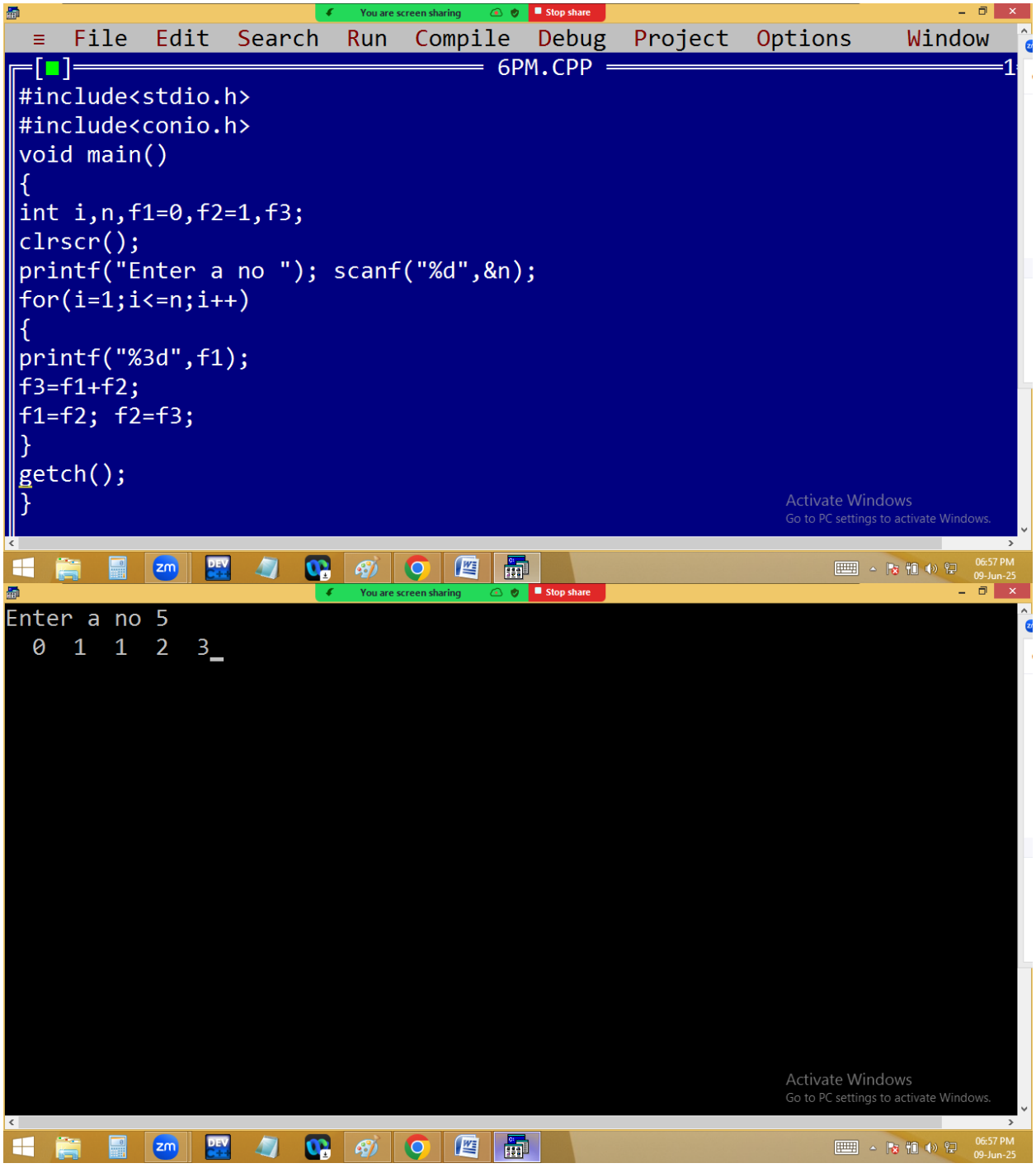

You are screen sharing
Stop share
Enter a no 3
Prime no
Activate Windows
Go to PC settings to activate Windows.
06:36 PM
09-Jun-25

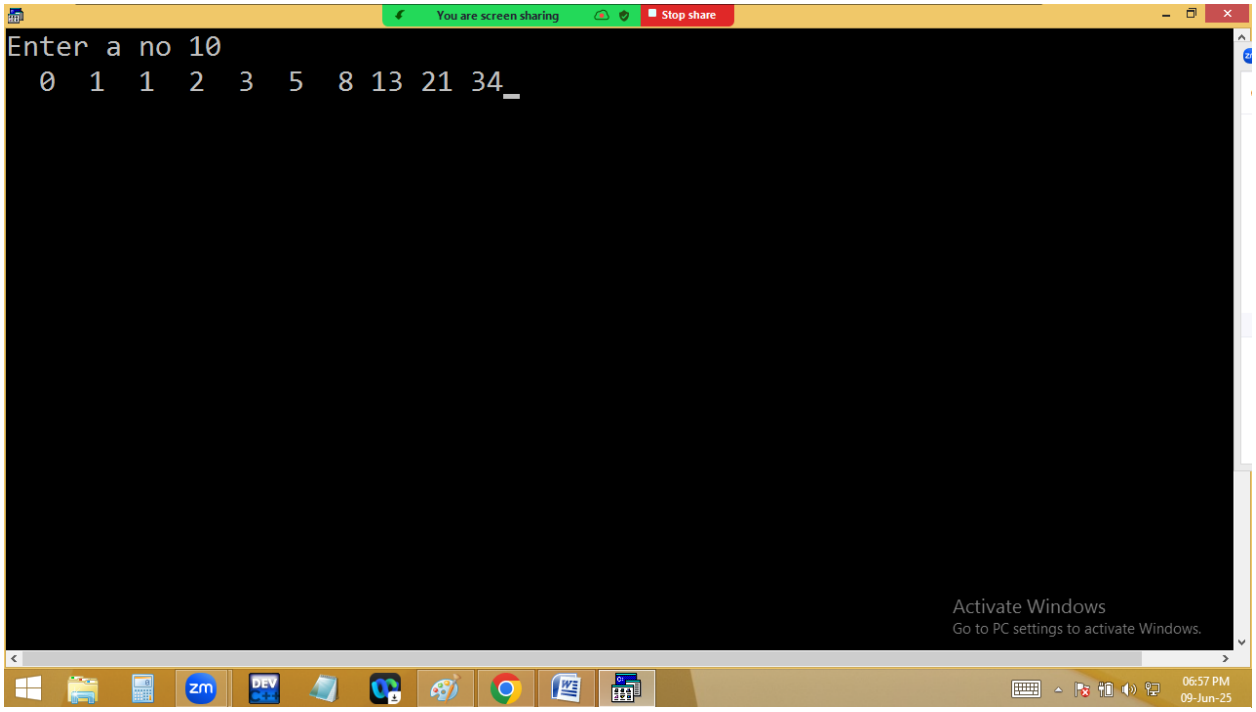




Fibonacci series:

N=5 → 0 1 1 2 3


You are screen sharing
Stop share
File Edit Search Run Compile Debug Project Options Window
6PM.CPP
#include<stdio.h>
#include<conio.h>
void main()
{
int i,n,f1=0,f2=1,f3;
clrscr();
printf("Enter a no "); scanf("%d",&n);
for(i=1;i<=n;i++)
{
printf("%3d",f1);
f3=f1+f2;
f1=f2; f2=f3;
}
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:57 PM
09-Jun-25
You are screen sharing
Stop share
Enter a no 5
0 1 1 2 3
Activate Windows
Go to PC settings to activate Windows.
06:57 PM
09-Jun-25



i	n	f1		f2	f3
1	5	0	+	1	1
2		1	+	1	2
3		1	+	2	3
4		2	+	3	5
5		3		5	

```
for( i=1;i<=5;i++ )
{
p(f1); 0 1 1 2 3
f3=f1+f2;
f1=f2;
f2=f3;
}
```

Finding Armstrong no or not?

1 is a single digit no → $1^1 = 1$

2 is a single digit no → $2^1 = 2$

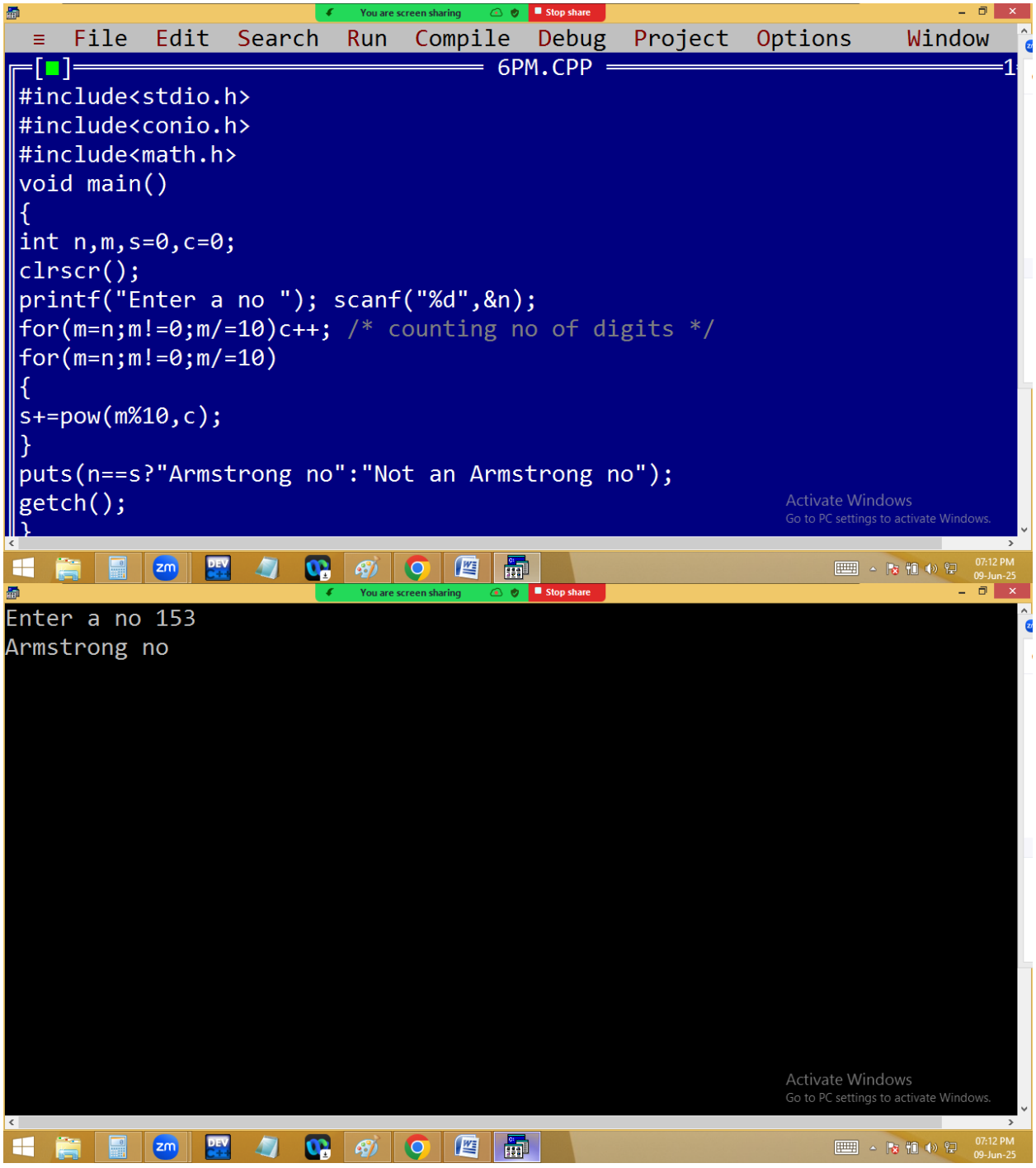
9 is a single digit no → $9^1 = 9$

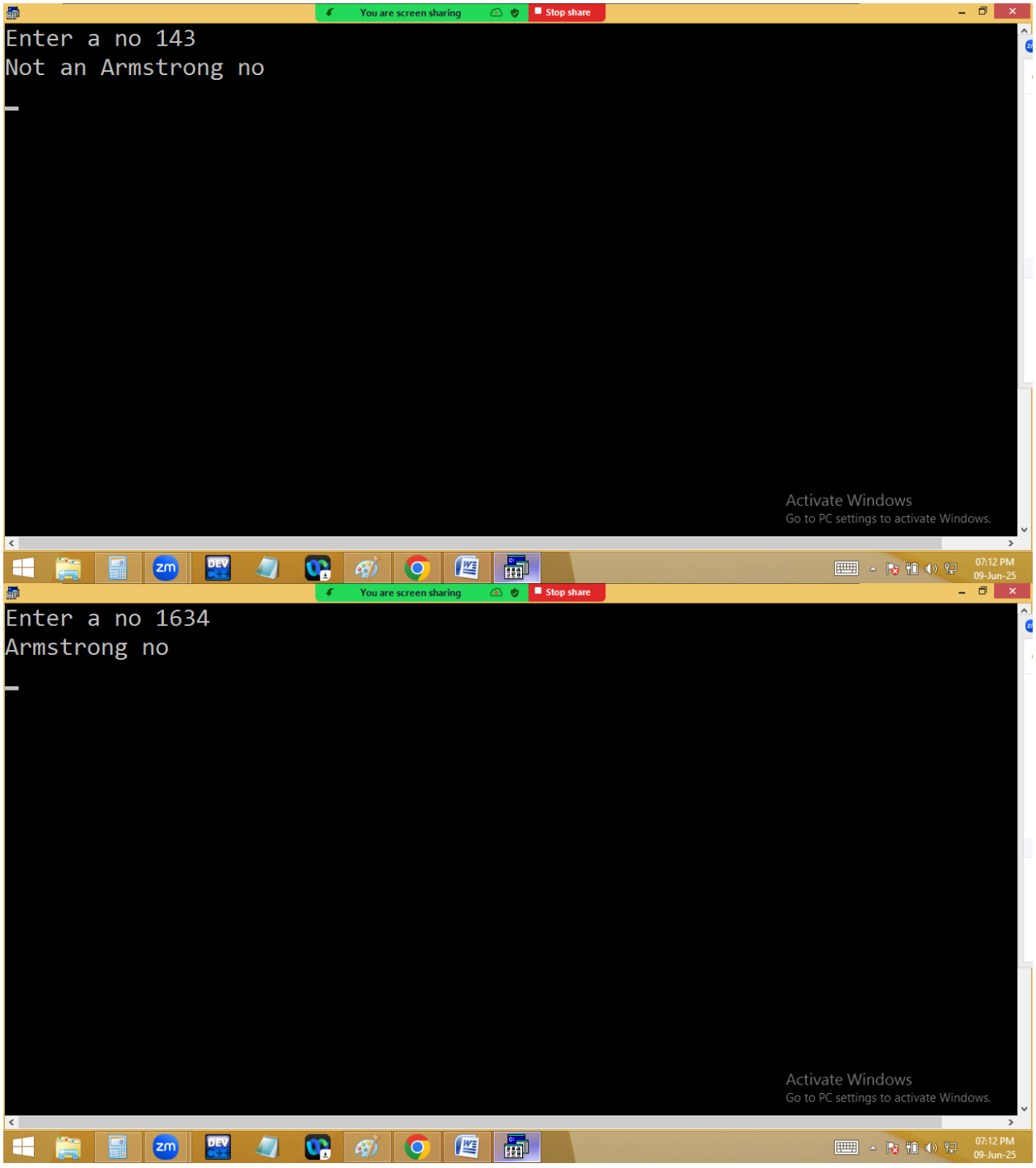
153 is a 3 digit no → $1^{3+}5^{3+}3^3 = 1+125+27=153$

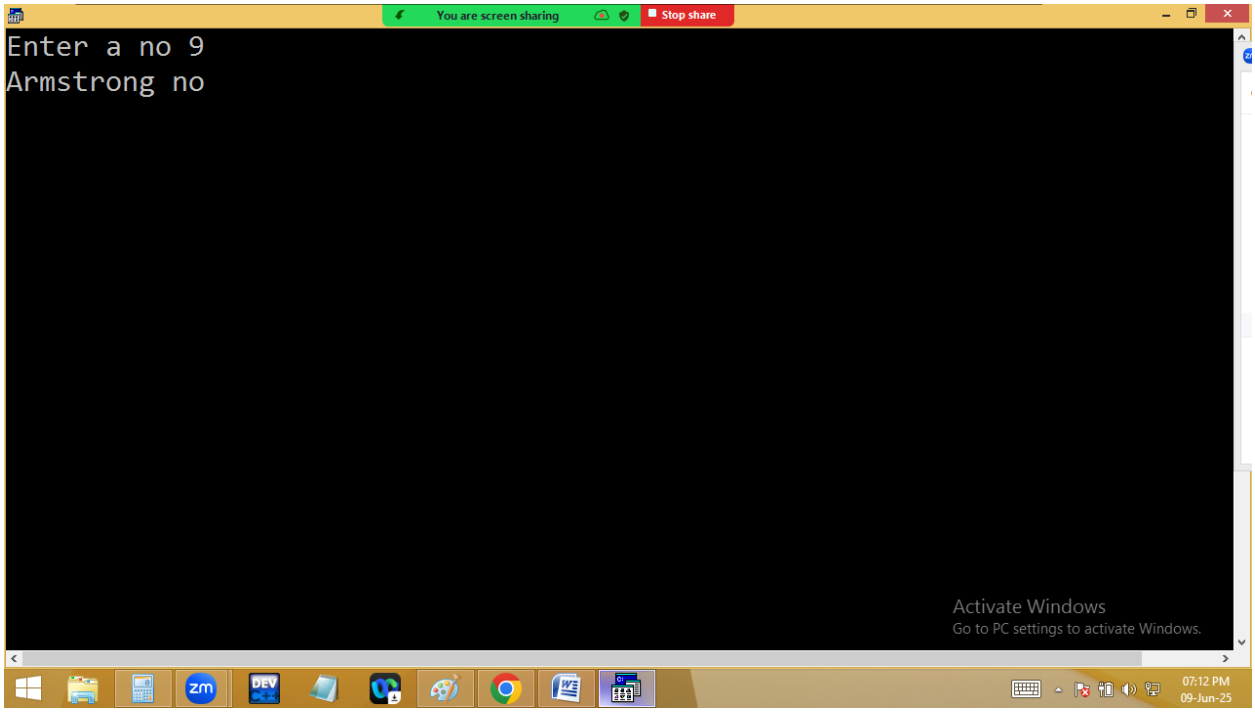
370, 371, 407, 1634, 8208,…

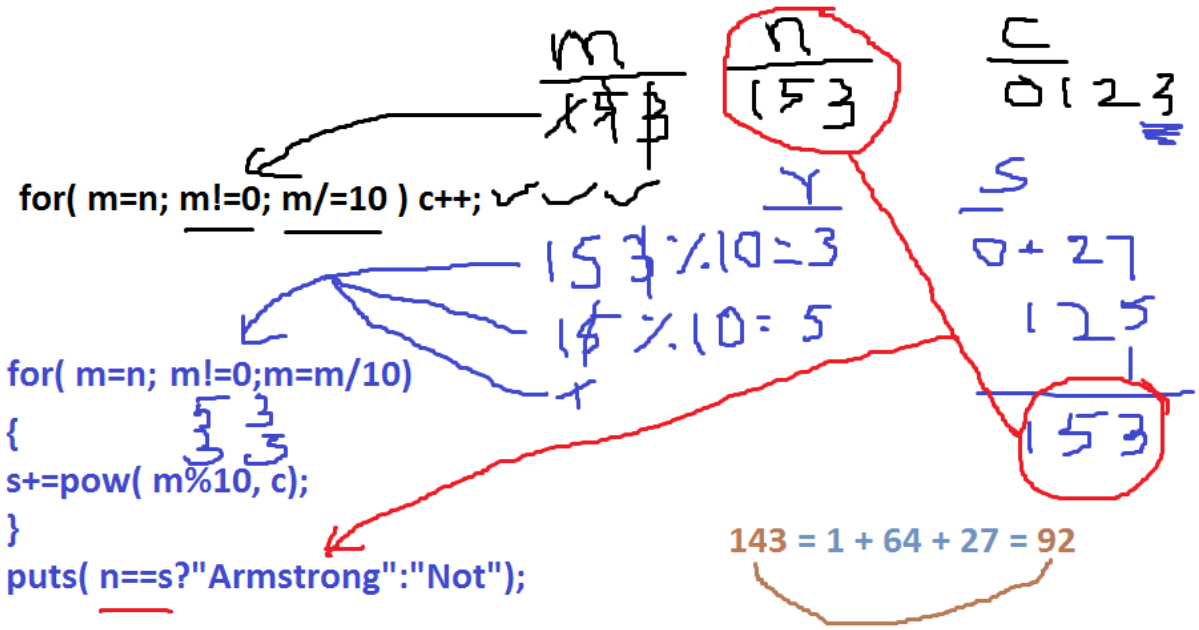
$1634 = 1^4 + 6^4+3^4+4^4 = 1634$

$8208 = 8^4 + 2^4 + 0^4 + 8^4 = 8208$

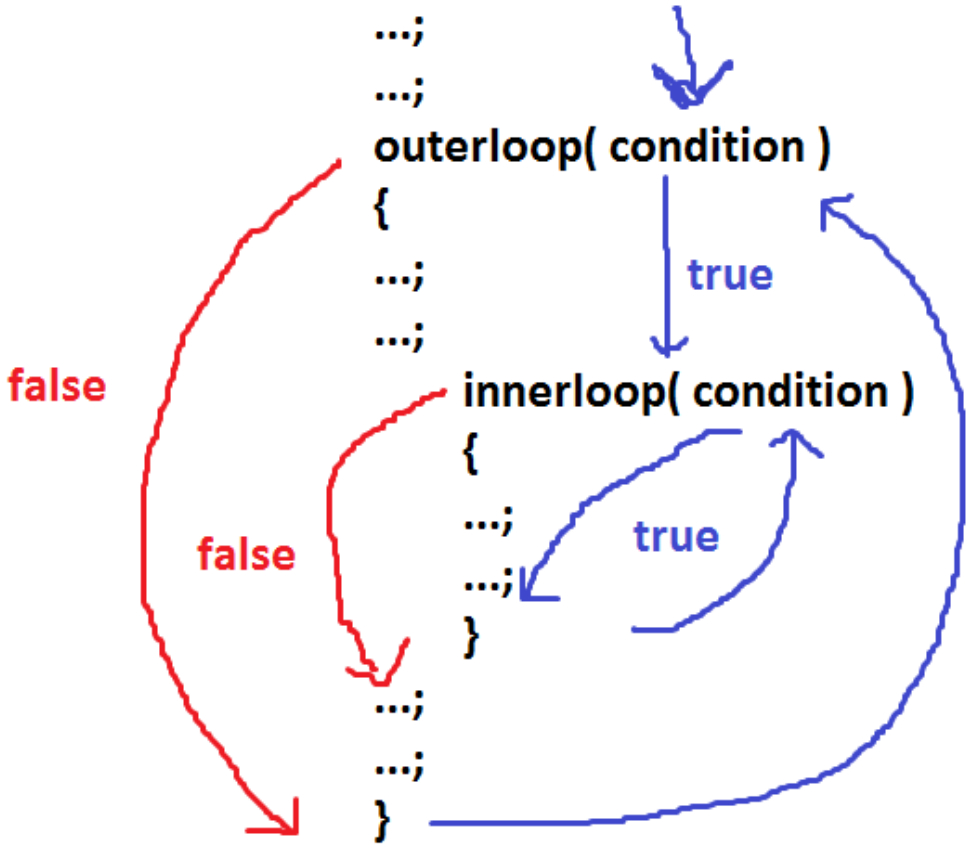

You are screen sharing
Stop share
File Edit Search Run Compile Debug Project Options Window
6PM.CPP
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
int n,m,s=0,c=0;
clrscr();
printf("Enter a no "); scanf("%d",&n);
for(m=n;m!=0;m/=10)c++; /* counting no of digits */
for(m=n;m!=0;m/=10)
{
s+=pow(m%10,c);
}
puts(n==s?"Armstrong no":"Not an Armstrong no");
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:12 PM
09-Jun-25
You are screen sharing
Stop share
Enter a no 153
Armstrong no
Activate Windows
Go to PC settings to activate Windows.
07:12 PM
09-Jun-25


You are screen sharing
Stop share
Enter a no 143
Not an Armstrong no
Activate Windows
Go to PC settings to activate Windows.
07:12 PM
09-Jun-25
You are screen sharing
Stop share
Enter a no 1634
Armstrong no
Activate Windows
Go to PC settings to activate Windows.
07:12 PM
09-Jun-25


You are screen sharing
Stop share
Enter a no 9
Armstrong no
Activate Windows
Go to PC settings to activate Windows.
07:12 PM
09-Jun-25


for(m=n; m!=0; m/=10) c++;
for(m=n; m!=0;m=m/10)
{
s+=pow(m%10, c);
}
puts(n==s?"Armstrong":"Not");
143 = 1 + 64 + 27 = 92

Nested loops: **Loop within loop is called nested loop.**

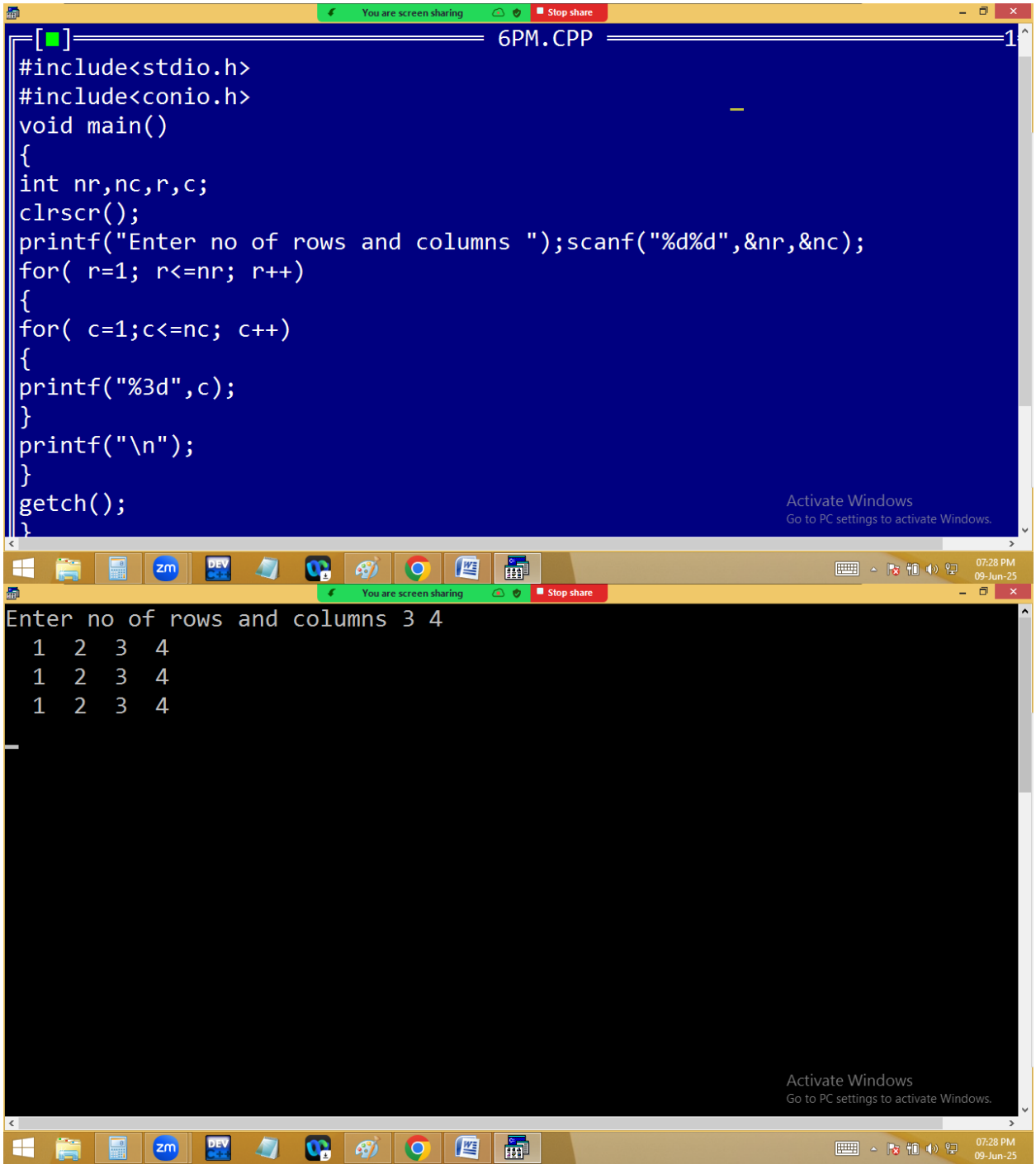


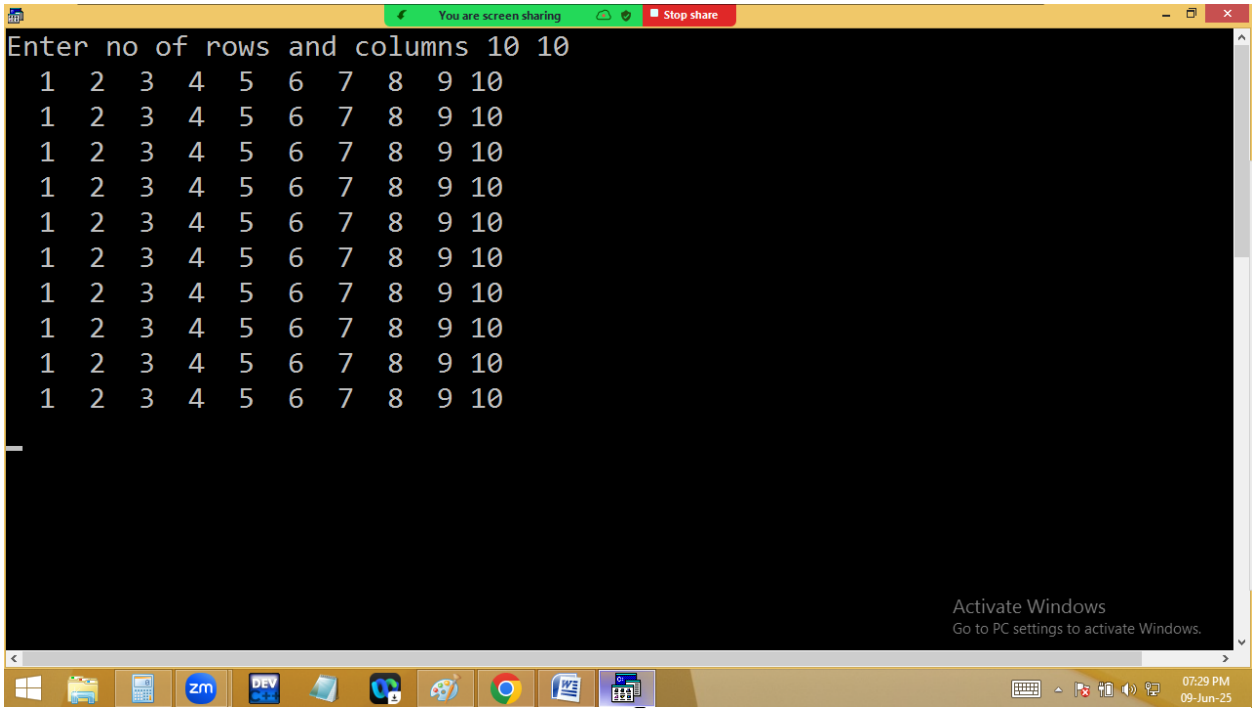
Printing below output?

1 2 3 4

1 2 3 4

1 2 3 4


You are screen sharing
Stop share
6PM.CPP
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
for(r=1; r<=nr; r++)
{
for(c=1;c<=nc; c++)
{
printf("%3d",c);
}
printf("\n");
}
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:28 PM
09-Jun-25
You are screen sharing
Stop share
Enter no of rows and columns 3 4
1 2 3 4
1 2 3 4
1 2 3 4
Activate Windows
Go to PC settings to activate Windows.
07:28 PM
09-Jun-25




```
for( r=1; r<=3; r++ )
{
for( c=1; c<=4; c++)
{
p( c );
}
p("\n");
}
```

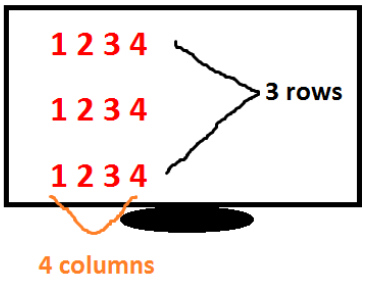
r = 1 to 4

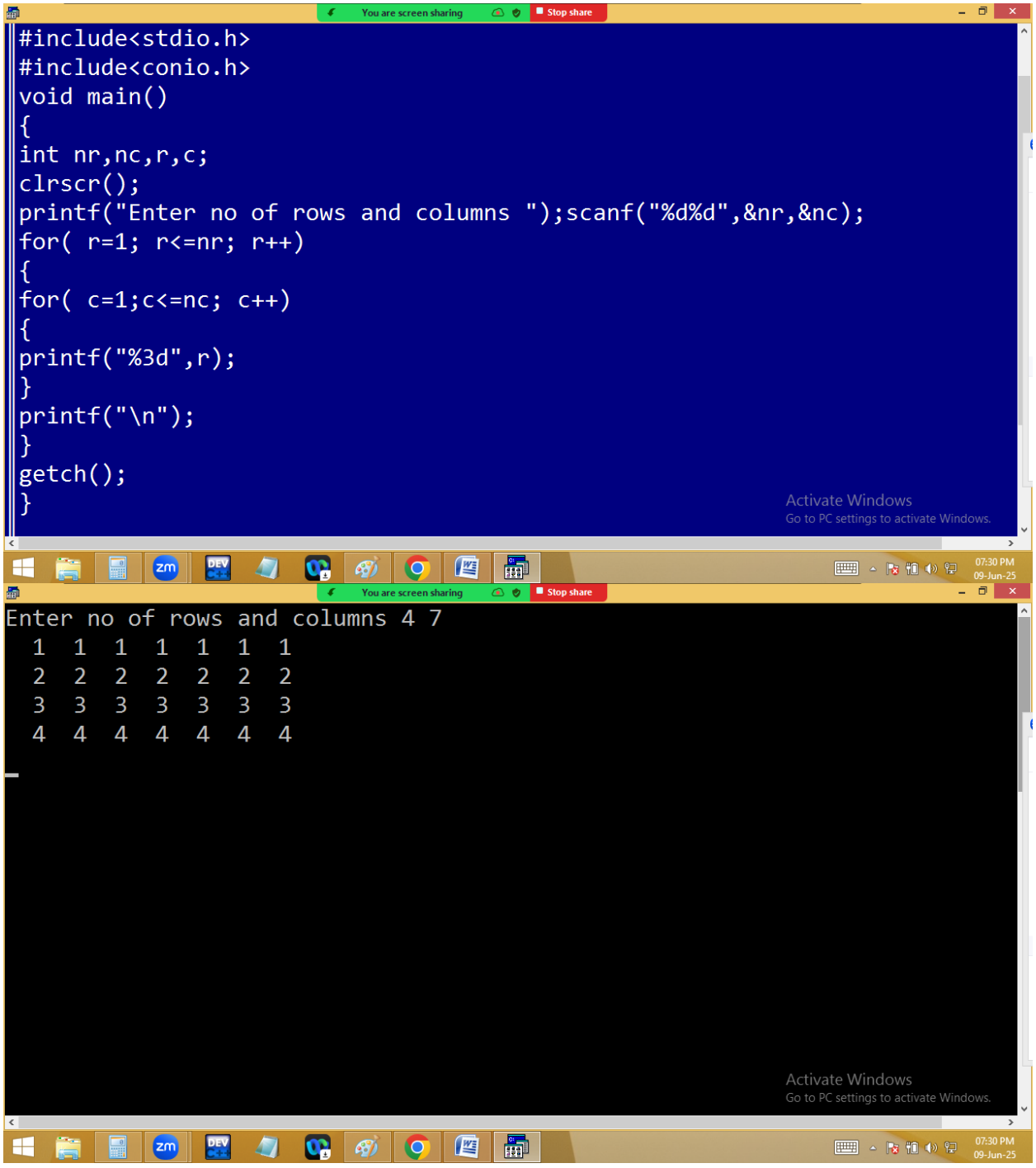
1 2 3 4

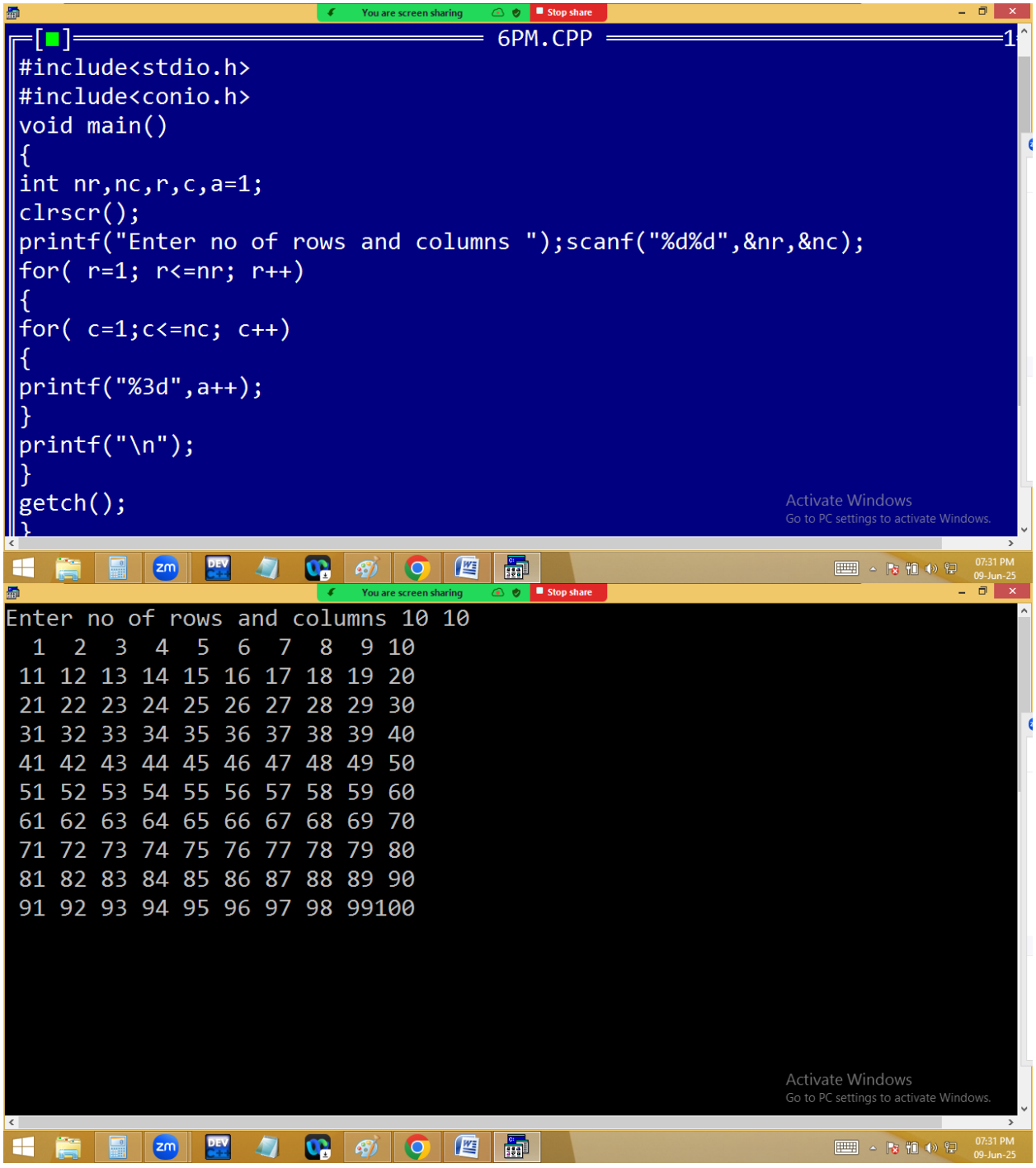
1 2 3 4

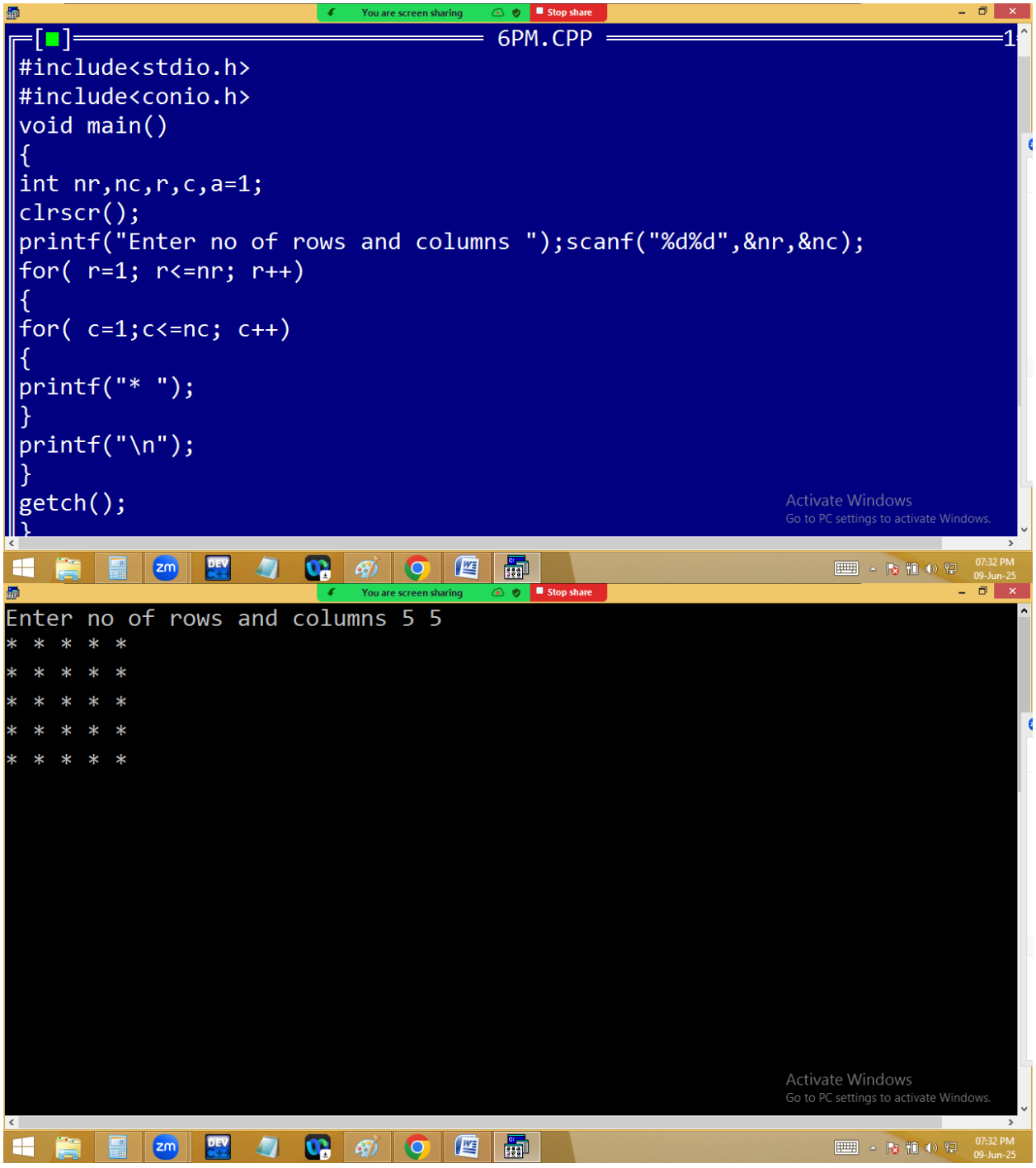
1 2 3 4

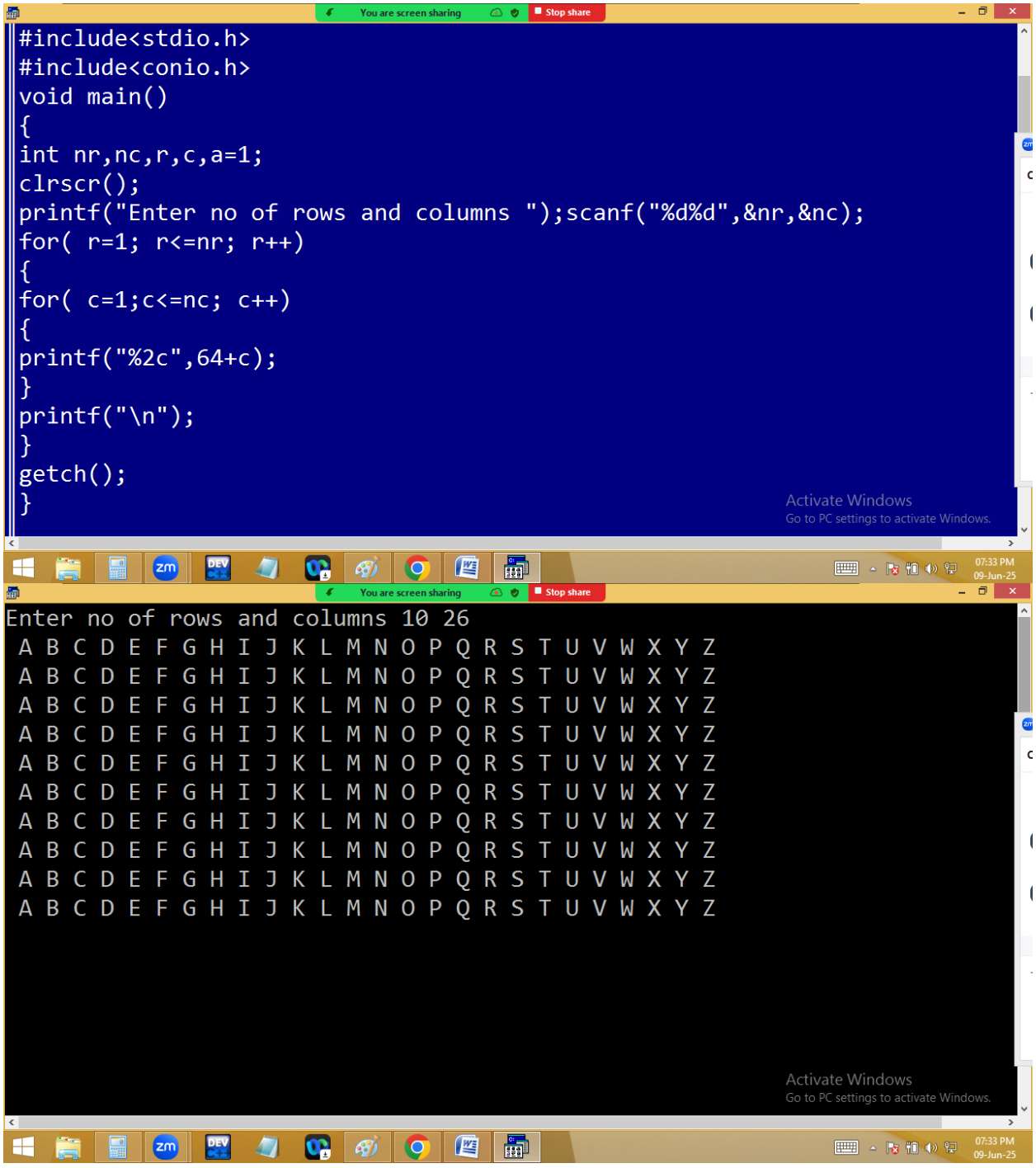
4

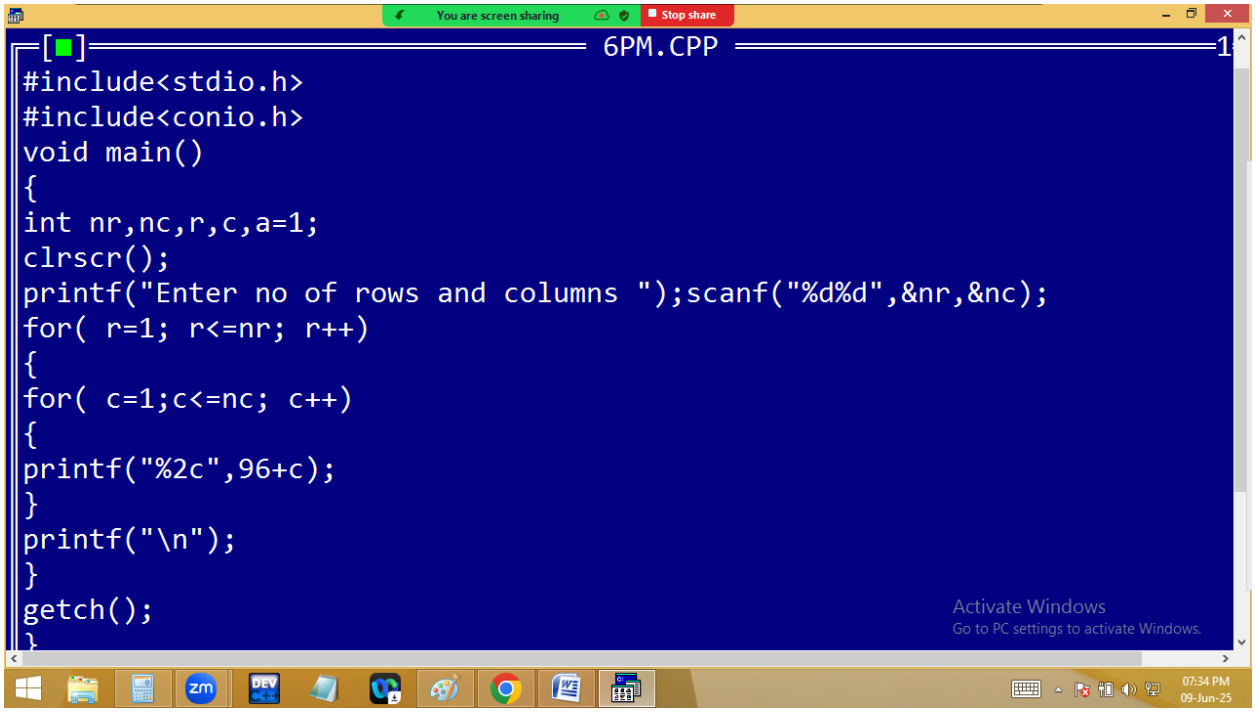



You are screen sharing
Stop share
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
for(r=1; r<=nr; r++)
{
for(c=1;c<=nc; c++)
{
printf("%3d",r);
}
printf("\n");
}
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:30 PM
09-Jun-25
You are screen sharing
Stop share
Enter no of rows and columns 4 7
1 1 1 1 1 1 1
2 2 2 2 2 2 2
3 3 3 3 3 3 3
4 4 4 4 4 4 4
Activate Windows
Go to PC settings to activate Windows.
07:30 PM
09-Jun-25


You are screen sharing
Stop share
6PM.CPP
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c,a=1;
clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
for(r=1; r<=nr; r++)
{
for(c=1;c<=nc; c++)
{
printf("%3d",a++);
}
printf("\n");
}
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:31 PM
09-Jun-25
You are screen sharing
Stop share
Enter no of rows and columns 10 10
1 2 3 4 5 6 7 8 9 10
11 12 13 14 15 16 17 18 19 20
21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40
41 42 43 44 45 46 47 48 49 50
51 52 53 54 55 56 57 58 59 60
61 62 63 64 65 66 67 68 69 70
71 72 73 74 75 76 77 78 79 80
81 82 83 84 85 86 87 88 89 90
91 92 93 94 95 96 97 98 99100
Activate Windows
Go to PC settings to activate Windows.
07:31 PM
09-Jun-25


You are screen sharing
Stop share
6PM.CPP
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c,a=1;
clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
for(r=1; r<=nr; r++)
{
for(c=1;c<=nc; c++)
{
printf("* ");
}
printf("\n");
}
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:32 PM
09-Jun-25
You are screen sharing
Stop share
Enter no of rows and columns 5 5
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
Activate Windows
Go to PC settings to activate Windows.
07:32 PM
09-Jun-25


You are screen sharing
Stop share
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,nc,r,c,a=1;
clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
for(r=1; r<=nr; r++)
{
for(c=1;c<=nc; c++)
{
printf("%2c",64+c);
}
printf("\n");
}
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:33 PM
09-Jun-25
You are screen sharing
Stop share
Enter no of rows and columns 10 26
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Activate Windows
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07:33 PM
09-Jun-25



Home work:

1.

abcd

ABCD

abcd

ABCD

2.

abcd

dcba

abcd

dcba

3.

AbCdEf

AbCdEf

AbCdEf